

Life in urban areas

Animals have had millions of years to adapt to earth's natural habitats, but only a fraction of that time to adjust to life in cities. Despite this, animals are never far away in built-up places. Their success is due mainly to "preadaptation"—characteristics that evolved to suit one way of life, or habitat, but that accidentally turn out to be useful for another. Thus, some animals thrive in man-made habitats that resemble the ones they would use in nature. Others succeed because they are highly adaptable and can exploit the opportunities that we inadvertently provide.

Feeding

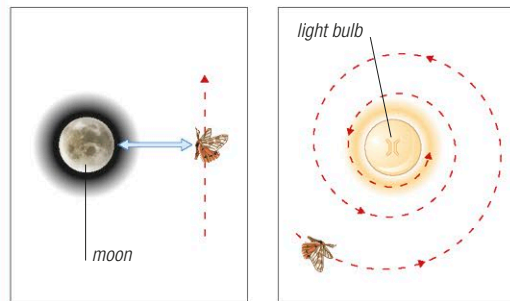
Some outdoor urban animals live on the same foods that they eat in the wild, but for scavenging species—such as raccoons, foxes, and pigeons—the daily fare is often very different than that of their natural homes. These versatile creatures will try any kind of leftover food, however unfamiliar it looks and smells, and this highly opportunistic streak is the secret of their great success. Modern food packaging can sometimes present problems, but they quickly learn how to tear or peck away at plastic and paper to get at the edible contents within.

Indoor animals get their food from one of 3 sources: the things we eat, the animals that eat those things, and the fabric of our homes. The first category contains a wide range of household pests, such as rats, mice, houseflies, and cockroaches; the second category consists chiefly of spiders, but also centipedes and geckos in warm parts of the world. Spiders are almost perfectly adapted to indoor life, and although widely disliked, they make a positive contribution by keeping indoor insect numbers in check. Animals in the third category are the least welcome of these uninvited guests. They include wood eaters such as termites and beetles, as well as insects

that attack other organic materials, such as wool. In many parts of the world, these animals are serious pests.

Light and warmth

In cities, streetlights light up the night sky, while heat from buildings and traffic makes them far warmer than nearby countryside. Artificial lights confuse insect navigation systems and interfere with birds' biological clocks. As a result, songbirds sometimes sing late at night, and some species start building nests in winter, convinced by the bright light that it is



DRAWN TO THE LIGHT

Moths navigate by maintaining a set angle to the moon, which, because it is so far away, works like a compass, enabling the moth to follow a straight line. With closer lights, such as street lamps, the "compass" appears to the moth to drift. The moth adjusts its path accordingly, and ends up spiraling inward.

DAWN RAID

Alert for signs of danger, and using its keen sense of smell and highly dexterous front paws, a raccoon sorts through trash from a container that it has overturned.



INTRODUCED SPECIES

Species that have set up home in unfamiliar parts of the world are plentiful in urban areas. Some have been introduced deliberately, while others arrive with imported food. Pets, also, are sometimes released, or manage to escape, into the wild. Some of these animals remain urban, but a few—such as the starling in North America—have gone on to colonize entire continents.



FAR FROM HOME

The rose-ringed parakeet from tropical Africa and southern Asia is good at dealing with cold, so it can thrive in parts of North America and Europe.

spring. Extra warmth is appreciated by a range of animals, from butterflies to birds. In some regions, starlings commute into cities on winter afternoons to roost on buildings where they are relatively warm.

Rooftop animals

For birds and bats, the tops of buildings can make ideal homes. High above the ground, and relatively undisturbed by people, animals feed and breed undisturbed. Some species nest in attics or under eaves, while others favor the tops of chimneys. Swifts, swallows, and martins are foremost among rooftop dwellers and are prime examples of preadaptation at work: they naturally nest on cliffs or in crevices, but the rapid spread of towns and cities has provided alternatives that have enabled them to extend their range to places where they would otherwise be rare.



LIFE ON THE ROOFTOPS

A pair of European white storks look down on a town from their chimney-top nest. Although they often nest on buildings, they feed in fields. They were once common urban visitors in northern Europe, but changing agricultural practices have reduced their numbers.

STRUCTURAL PESTS

Since humans first started to build wood houses, wood-eating insects have been a problem. Wood-eating beetles are common in temperate regions, while termites attack timber in the tropics. At one time, little could be done to control severe attacks; today, insecticides are effective at keeping them in check. Even so, many of these animals, especially wood-boring beetles, are far more widespread than they were because they have been exported in the lumber shipped all over the world.



EATEN FROM INSIDE

Deathwatch beetle larvae chew tunnels through old lumber. During the 5 years they take to turn into adults, they can weaken lumber so much that it eventually collapses. A related species attacks furniture wood.